CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An ant sting is treated with baking soda because: 1

(a) soda dissolves the sting
(b) baking soda is an acid
(c) acid in the sting is neutralised by a mild base
(d) soda cools the area

2. Assertion (A): A solution of NaCl is neutral. Reason (R): NaCl is a salt of strong acid and strong base. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Mass of baking soda needed to give 22 g of CO_2 on heating (NaHCO3 = 84, CO_2 = 44) is: 1

(a) 84 g **(b)** 168 g **(c)** 44 g **(d)** 42 g

4. Milk of magnesia is suitable as an antacid because: 1

(a) it is a strong acid
(b) it is a salt
(c) it is an indicator
(d) it is a mild base that neutralises the excess acid

5. Assertion (A): A solution with pH 3 is 10 times more acidic than a solution with pH 4. Reason (R): A fall of one pH unit means a ten-fold increase in H^+ concentration. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A solution of pH 2 is mixed with equal volume of a solution of pH 12 (same normality). The mixture is: 1

(a) acidic **(b)** cannot be said **(c)** neutral **(d)** basic

7. Compound X is used for making cakes soft. On adding dilute HCl it gives brisk effervescence. X is: 1

(a) common salt **(b)** gypsum **(c)** baking soda **(d)** plaster of Paris

8. Assertion (A): Hydrochloric acid is stronger than acetic acid at the same concentration. Reason (R): HCl and acetic acid both contain hydrogen. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. How many times is the H^+ concentration in blood (pH 7.4) greater than in a solution of pH 8.4? 1

(a) 10 **(b)** 2 **(c)** 100 **(d)** 1

10. Sodium hydrogencarbonate in a soda-acid extinguisher reacts with: 1

(a) sodium chloride **(b)** water alone **(c)** sodium hydroxide **(d)** sulphuric acid

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is Na2CO3 solution more basic than NaHCO3 solution? 2

12. Calculate the mass of water of crystallisation in 286 g of washing soda (Na = 23, C = 12, O = 16, H = 1). 2

13. Why does an aqueous solution of anhydrous copper sulphate turn blue? 2

14. An unknown white solid evolves CO_2 with acid and is stable on heating. Is it baking soda? Justify. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 2.1 g of baking soda is heated completely. Find the mass of CO_2 released (NaHCO3 = 84, CO_2 = 44). 3

16. Which gas would you identify from the 3
(i) pop sound,
(ii) milky lime water, and what reaction produces each from dilute HCl?

17. Explain why distilled water used for a pH test shows pH 7 but tap water may vary. 3

OR

Explain why the mixture of an acid and a base is not always neutral when mixed in equal volumes. 3

18. Brine is electrolysed. State what happens at each electrode and why the solution becomes basic. 3

OR

Explain what happens when bleaching powder is left open in air. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Solutions A-E have pH 1, 3, 7, 9 and 13. Identify the best antacid, one turning phenolphthalein pink, one for stomach acid, and compare H^+ of A and C. 5

OR

Describe an experiment to prove that the gas formed on heating baking soda is carbon dioxide and that a base is left. 5

20. Explain how the chlor-alkali products are the starting materials for other chemicals, with equations including bleaching powder and baking soda. 5

OR

Why is it not possible to prepare plaster of Paris by heating gypsum at 473 K? What would form and what are the consequences? 5

END OF QUESTION PAPER — **Total: 40 Marks**